

Question Number	Answer	Additional Guidance	Mark
3ai	An explanation that includes the following: • prevents escape / spreading of (bacteria / <i>E.coli</i>) (1) • prevents {contamination of the agar or cultures with (other types of) bacteria) / cross contamination /effect on the results} (1)	accept ref validity If no other marks awarded ACCEPT prevents contamination	Expert 2

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3a ii	An answer that includes four of the following points: • description of how a bacterial culture is made (1) • description of addition of extract (1) • control of feature of independent variable described (1) • incubate {at suitable quoted temperature / for suitable quoted time / same length of time} (1) • statement of what is measured related to method above if awarded (1)	e.g. paper disc / cut well e.g. disc size / wells same size / {volume / age} / of extract temp. <30 but > 10, time 1 day to 7 days zone of inhibition clear area / turbidity	expert 4

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3b	A table with the following features: - suitable table drawn (1) - all headings correct with units (1) - all data correctly entered (1)	Must all be quoted to one decimal place. <table border="1"> <thead> <tr> <th colspan="4">Inhibition of growth / au</th></tr> <tr> <th colspan="2">Roselle</th><th colspan="2">Clove</th></tr> <tr> <th>Ethanol</th><th>Water</th><th>Ethanol</th><th>Water</th></tr> </thead> <tbody> <tr> <td>21.1 / 21.0</td><td>15.6 / 15.5</td><td>17.4 / 17.5</td><td>13.0/ 13.1 / 13.2</td></tr> </tbody> </table> Units in cells loses mp 2	Inhibition of growth / au				Roselle		Clove		Ethanol	Water	Ethanol	Water	21.1 / 21.0	15.6 / 15.5	17.4 / 17.5	13.0/ 13.1 / 13.2	expert 3
Inhibition of growth / au																			
Roselle		Clove																	
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3ci	1.2:1 / 1.23:1 / 0.83:1	ACCEPT the other way round in any case	Grad 1

Question Number	Answer	Additional Guidance	Mark
3cii	An answer that includes the following points: - Ethanol extract {is more effective / better than / gives higher inhibition of growth than / has higher antimicrobial properties} (extract made with water) (1) - Roselle (extract) {is more effective / better / gives higher inhibition of growth than / has higher	ACCEPT reverse argument	expert 2

	antimicrobial properties} } (than the clove extract) (1)		
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3d	An answer that includes two of the following points: • {enzymes / protein} needed for growth / relevant processes described (1) • pH is no longer optimal (for growth) (1) • because {enzymes / proteins} are denatured (by pH change) (1) • lower pH affects (hydrogen / ionic) bonding / R groups (in proteins / enzymes) (1)		expert 2